```
clc;
clear;
close all;

%% ============================================================
% DESIGN AND COMPARATIVE ANALYSIS OF
% BUTTERWORTH, CHEBYSHEV TYPE-I AND ELLIPTIC IIR FILTERS
%% ============================================================

%% 1. INPUT PARAMETERS

Fs  = 1000;     % Sampling frequency (Hz)
Fp  = 100;      % Passband edge frequency (Hz)
Fst = 150;      % Stopband edge frequency (Hz)
Rp  = 1;        % Passband ripple (dB)
Rs  = 40;       % Stopband attenuation (dB)

% Normalized frequencies
Wp  = Fp/(Fs/2);
Wst = Fst/(Fs/2);

%% 2. CHEBYSHEV TYPE-I THEORETICAL FORMULA

epsilon = sqrt(10^(Rp/10) - 1);

N_formula = acosh(sqrt((10^(Rs/10) - 1) / ...
                       (10^(Rp/10) - 1))) / ...
            acosh(Fst/Fp);

N_theory = ceil(N_formula);

%% 3. BUTTERWORTH FILTER DESIGN

[Nb, Wb] = buttord(Wp, Wst, Rp, Rs);
[bB, aB] = butter(Nb, Wb, 'low');

%% 4. CHEBYSHEV TYPE-I FILTER DESIGN

[Nc, Wc] = cheb1ord(Wp, Wst, Rp, Rs);
[bC, aC] = cheby1(Nc, Rp, Wc, 'low');

%% 5. ELLIPTIC FILTER DESIGN

[Ne, We] = ellipord(Wp, Wst, Rp, Rs);
[bE, aE] = ellip(Ne, Rp, Rs, We, 'low');

%% 6. DISPLAY INPUT PARAMETERS
```

```
fprintf('\n==================================================\n');
```

```
==================================================
```

```
fprintf('          IIR FILTER COMPARATIVE ANALYSIS\n');
```

```
          IIR FILTER COMPARATIVE ANALYSIS
```

```
fprintf('==================================================\n');
```

```
==================================================
```

```
fprintf('\nINPUT PARAMETERS\n');
```

```
INPUT PARAMETERS
```

```
fprintf('Sampling Frequency       = %.2f Hz\n', Fs);
```

```
Sampling Frequency       = 1000.00 Hz
```

```
fprintf('Passband Edge Frequency  = %.2f Hz\n', Fp);
```

```
Passband Edge Frequency  = 100.00 Hz
```

```
fprintf('Stopband Edge Frequency  = %.2f Hz\n', Fst);
```

```
Stopband Edge Frequency  = 150.00 Hz
```

```
fprintf('Passband Ripple          = %.2f dB\n', Rp);
```

```
Passband Ripple          = 1.00 dB
```

```
fprintf('Stopband Attenuation     = %.2f dB\n', Rs);
```

```
Stopband Attenuation     = 40.00 dB
```

```
%% 7. CHEBYSHEV TYPE-I FORMULA RESULTS

fprintf('\nCHEBYSHEV TYPE-I FORMULA\n');
```

```
CHEBYSHEV TYPE-I FORMULA
```

```
fprintf('Ripple Factor epsilon    = %.4f\n', epsilon);
```

```
Ripple Factor epsilon    = 0.5088
```

```
fprintf('Calculated Order         = %.4f\n', N_formula);
```

```
Calculated Order         = 6.2071
```

```
fprintf('Theoretical Minimum N    = %d\n', N_theory);
```

```
Theoretical Minimum N    = 7
```

```
%% 8. FILTER ORDER COMPARISON
```

```
fprintf('\n=====================================================\n');
```

```
=====================================================
```

```
fprintf('              FILTER ORDER COMPARISON\n');
```

```
              FILTER ORDER COMPARISON
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```

```
fprintf('Butterworth Order        = %d\n', Nb);
```

```
Butterworth Order        = 12
```

```
fprintf('Chebyshev Type-I Order   = %d\n', Nc);
```

```
Chebyshev Type-I Order   = 6
```

```
fprintf('Elliptic Order           = %d\n', Ne);
```

```
Elliptic Order           = 4
```

```
%% 9. DISPLAY FILTER COEFFICIENTS

fprintf('\n=====================================================\n');
```

```
=====================================================
```

```
fprintf('           BUTTERWORTH COEFFICIENTS\n');
```

```
           BUTTERWORTH COEFFICIENTS
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```

```
fprintf('Numerator bB =\n');
```

```
Numerator bB =
```

```
disp(bB);
```

```
   1.0e-03 *

    0.0002    0.0027    0.0146    0.0488    0.1098    0.1757    0.2050    0.1757    0.1098    0.0488    0.
```

```
fprintf('Denominator aB =\n');
```

```
Denominator aB =
```

```
disp(aB);
```

```
    1.0000   -6.8844   22.4341  -45.5100   63.7725  -64.8501   48.9643  -27.6092   11.5218   -3.4662    0.
```

```
fprintf('\n=====================================================\n');
```

```
=====================================================
```

```
fprintf('          CHEBYSHEV TYPE-I COEFFICIENTS\n');
```

```
          CHEBYSHEV TYPE-I COEFFICIENTS
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```

```
fprintf('Numerator bC =\n');
```

```
Numerator bC =
```

```
disp(bC);
```

```
   1.0e-03 *

    0.0464    0.2782    0.6956    0.9274    0.6956    0.2782    0.0464
```

```
fprintf('Denominator aC =\n');
```

```
Denominator aC =
```

```
disp(aC);
```

```
    1.0000   -4.8694   10.3807  -12.3367    8.5969   -3.3268    0.5586
```

```
fprintf('\n=====================================================\n');
```

```
=====================================================
```

```
fprintf('             ELLIPTIC COEFFICIENTS\n');
```

```
             ELLIPTIC COEFFICIENTS
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```

```
fprintf('Numerator bE =\n');
```

```
Numerator bE =
```

```
disp(bE);
```

```
    0.0197   -0.0171    0.0333   -0.0171    0.0197
```

```
fprintf('Denominator aE =\n');
```

```
Denominator aE =
```

```
disp(aE);
```

```
    1.0000   -3.0330    3.8118   -2.2911    0.5554
```

```
%% 10. FREQUENCY RESPONSES

N = 4096;

[HB, f] = freqz(bB, aB, N, Fs);
[HC, ~] = freqz(bC, aC, N, Fs);
[HE, ~] = freqz(bE, aE, N, Fs);

% Magnitude responses in dB
MB = 20*log10(abs(HB) + eps);
MC = 20*log10(abs(HC) + eps);
ME = 20*log10(abs(HE) + eps);

%% 11. PASSBAND RIPPLE CALCULATION

P = f <= Fp;

Ripple_B = max(MB(P)) - min(MB(P));
Ripple_C = max(MC(P)) - min(MC(P));
Ripple_E = max(ME(P)) - min(ME(P));

%% 12. STOPBAND ATTENUATION CALCULATION

S = f >= Fst;

Stop_B = -max(MB(S));
Stop_C = -max(MC(S));
Stop_E = -max(ME(S));

%% 13. PERCENTAGE DEVIATION CALCULATION

Dev_C_Ripple = abs(Ripple_C - Rp)/Rp * 100;

Dev_B_Stop = abs(Stop_B - Rs)/Rs * 100;
Dev_C_Stop = abs(Stop_C - Rs)/Rs * 100;
Dev_E_Stop = abs(Stop_E - Rs)/Rs * 100;

%% 14. TRANSITION BANDWIDTH

TransitionBW = Fst - Fp;

%% 15. DISPLAY FILTER PERFORMANCE
```

```
fprintf('\n=====================================================\n');
```

```
=====================================================
```

```
fprintf('                FILTER PERFORMANCE\n');
```

```
                FILTER PERFORMANCE
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```

```
fprintf('\nPASSBAND RIPPLE\n');
```

```
PASSBAND RIPPLE
```

```
fprintf('Butterworth      = %.4f dB\n', Ripple_B);
```

```
Butterworth      = 0.8033 dB
```

```
fprintf('Chebyshev Type-I = %.4f dB\n', Ripple_C);
```

```
Chebyshev Type-I = 1.0000 dB
```

```
fprintf('Elliptic         = %.4f dB\n', Ripple_E);
```

```
Elliptic         = 1.0000 dB
```

```
fprintf('\nSTOPBAND ATTENUATION\n');
```

```
STOPBAND ATTENUATION
```

```
fprintf('Butterworth      = %.4f dB\n', Stop_B);
```

```
Butterworth      = 40.0198 dB
```

```
fprintf('Chebyshev Type-I = %.4f dB\n', Stop_C);
```

```
Chebyshev Type-I = 41.3365 dB
```

```
fprintf('Elliptic         = %.4f dB\n', Stop_E);
```

```
Elliptic         = 40.0000 dB
```

```
fprintf('\nCHEBYSHEV TYPE-I RIPPLE DEVIATION\n');
```

```
CHEBYSHEV TYPE-I RIPPLE DEVIATION
```

```
fprintf('Specified Ripple       = %.2f dB\n', Rp);
```

```
Specified Ripple       = 1.00 dB
```

```
fprintf('Measured Ripple        = %.4f dB\n', Ripple_C);
```

```
Measured Ripple        = 1.0000 dB
```

```
fprintf('Percentage Deviation   = %.4f %%\n', Dev_C_Ripple);
```

```
Percentage Deviation   = 0.0000 %
```

```
fprintf('\nSTOPBAND ATTENUATION DEVIATION\n');
```

```
STOPBAND ATTENUATION DEVIATION
```

```
fprintf('Butterworth Deviation  = %.4f %%\n', Dev_B_Stop);
```

```
Butterworth Deviation  = 0.0494 %
```

```
fprintf('Chebyshev-I Deviation  = %.4f %%\n', Dev_C_Stop);
```

```
Chebyshev-I Deviation  = 3.3413 %
```

```
fprintf('Elliptic Deviation     = %.4f %%\n', Dev_E_Stop);
```

```
Elliptic Deviation     = 0.0000 %
```

```
fprintf('\nTRANSITION CHARACTERISTICS\n');
```

```
TRANSITION CHARACTERISTICS
```

```
fprintf('Passband Edge          = %.2f Hz\n', Fp);
```

```
Passband Edge          = 100.00 Hz
```

```
fprintf('Stopband Edge          = %.2f Hz\n', Fst);
```

```
Stopband Edge          = 150.00 Hz
```

```
fprintf('Transition Bandwidth   = %.2f Hz\n', TransitionBW);
```

```
Transition Bandwidth   = 50.00 Hz
```

```
%% 16. MAGNITUDE RESPONSE COMPARISON

figure;

plot(f, MB, 'LineWidth', 1.5);
hold on;
plot(f, MC, 'LineWidth', 1.5);
plot(f, ME, 'LineWidth', 1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Comparison of IIR Filter Magnitude Responses');

legend('Butterworth', 'Chebyshev Type-I', 'Elliptic', ...
```

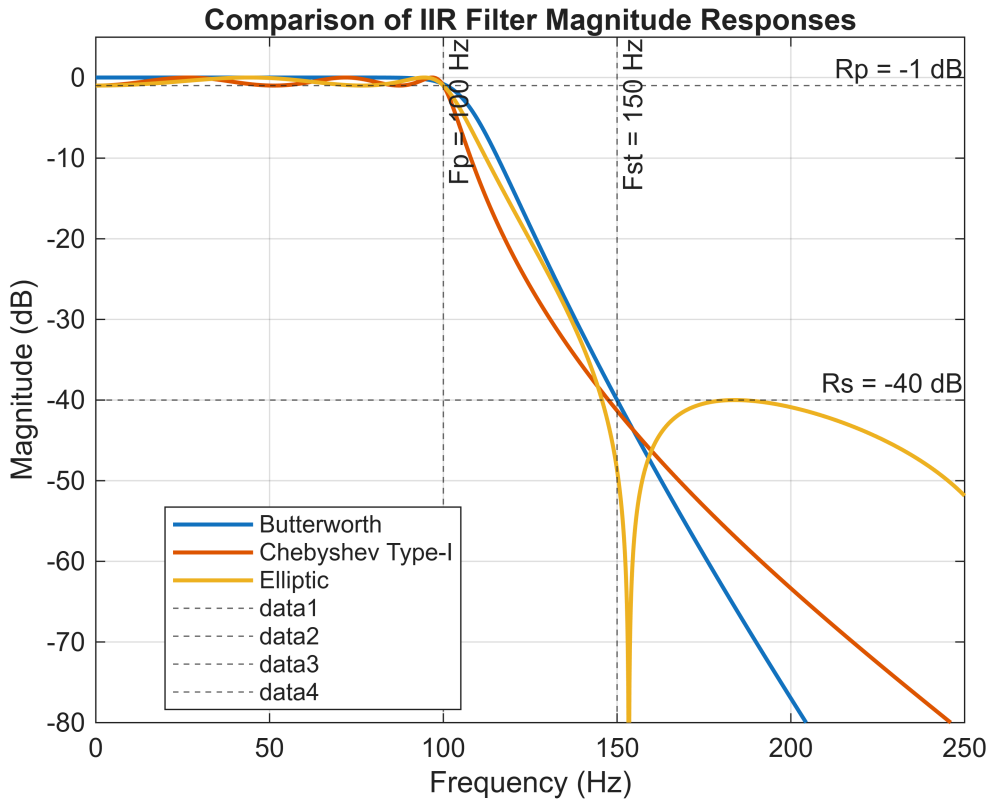
```
       'Location', 'best');

xlim([0 250]);
ylim([-80 5]);

xline(Fp, '--', 'Fp = 100 Hz');
xline(Fst, '--', 'Fst = 150 Hz');

yline(-Rp, '--', 'Rp = -1 dB');
yline(-Rs, '--', 'Rs = -40 dB');

hold off;
```




```
%% 17. PHASE RESPONSE COMPARISON

Phase_B = unwrap(angle(HB)) * 180/pi;
Phase_C = unwrap(angle(HC)) * 180/pi;
Phase_E = unwrap(angle(HE)) * 180/pi;

figure;

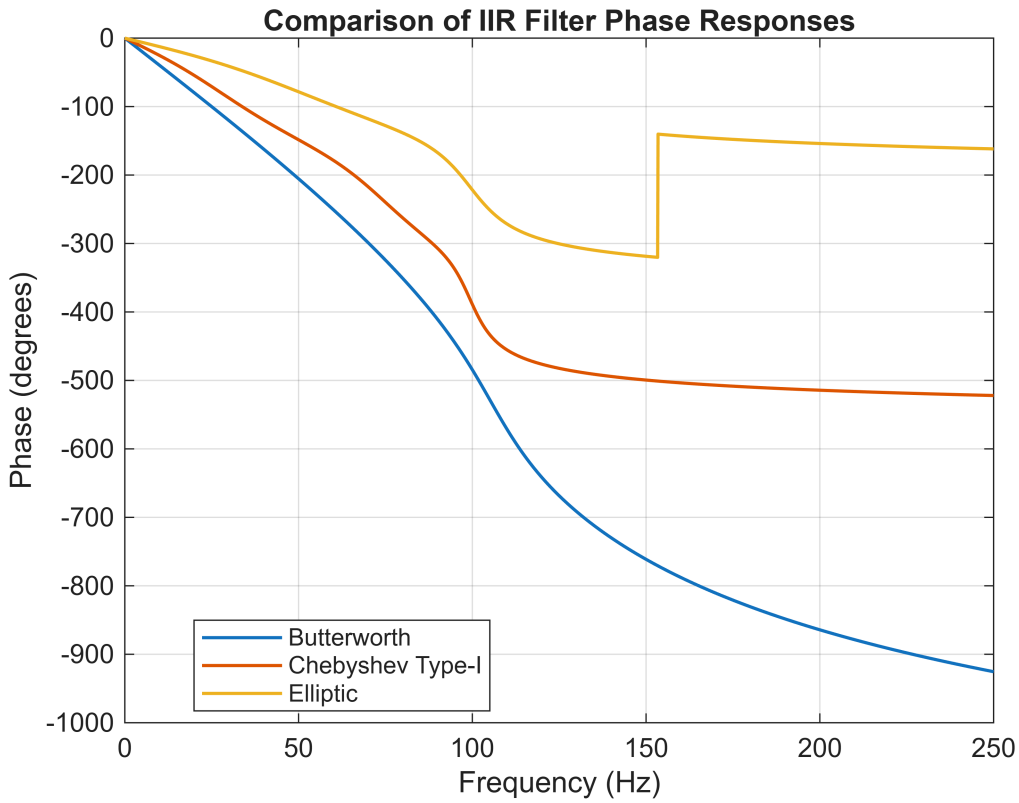
plot(f, Phase_B, 'LineWidth', 1.2);
hold on;
plot(f, Phase_C, 'LineWidth', 1.2);
plot(f, Phase_E, 'LineWidth', 1.2);
```

```
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('Comparison of IIR Filter Phase Responses');

legend('Butterworth', 'Chebyshev Type-I', 'Elliptic', ...
       'Location', 'best');

xlim([0 250]);

hold off;
```




```
%% 18. FILTER ORDER COMPARISON GRAPH

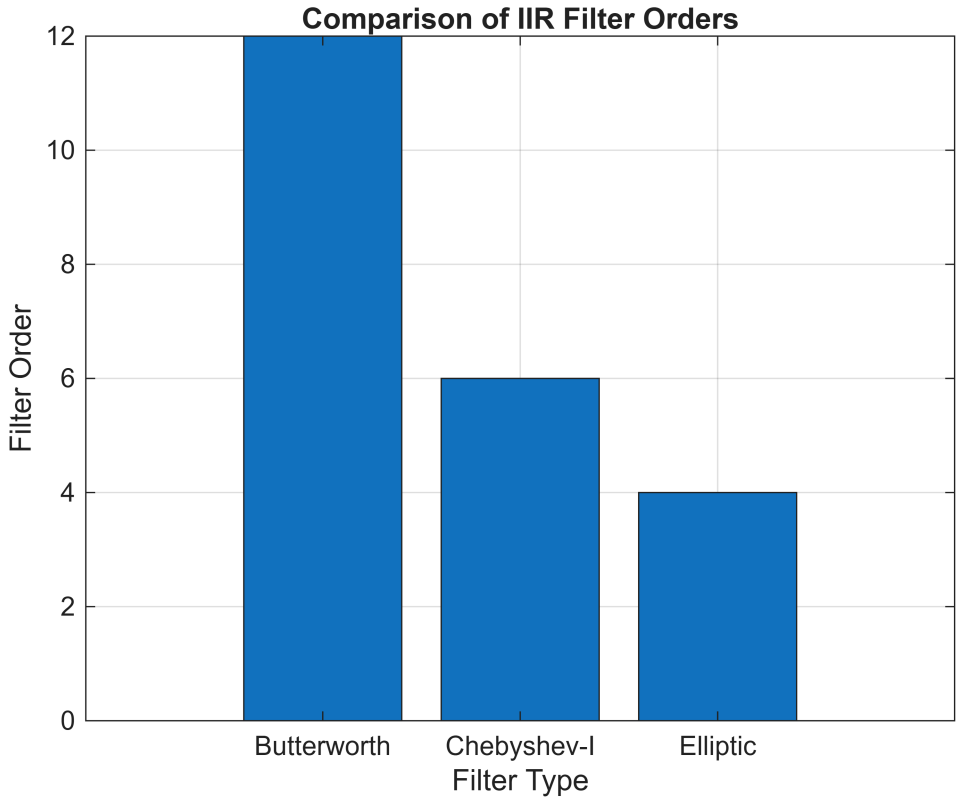
figure;

bar([Nb, Nc, Ne]);

set(gca, 'XTick', 1:3);
set(gca, 'XTickLabel', ...
    {'Butterworth', 'Chebyshev-I', 'Elliptic'});

ylabel('Filter Order');
xlabel('Filter Type');
title('Comparison of IIR Filter Orders');
```

```
grid on;
```


Comparison of IIR Filter Orders
Filter Order
0
2
4
6
8
10
12
Butterworth
Chebyshev-I
Elliptic
Filter Type